בקובץ פלט לתרגיל מספר 3

שאלה 3

ביקורות חיוביות:

פרמטרים :

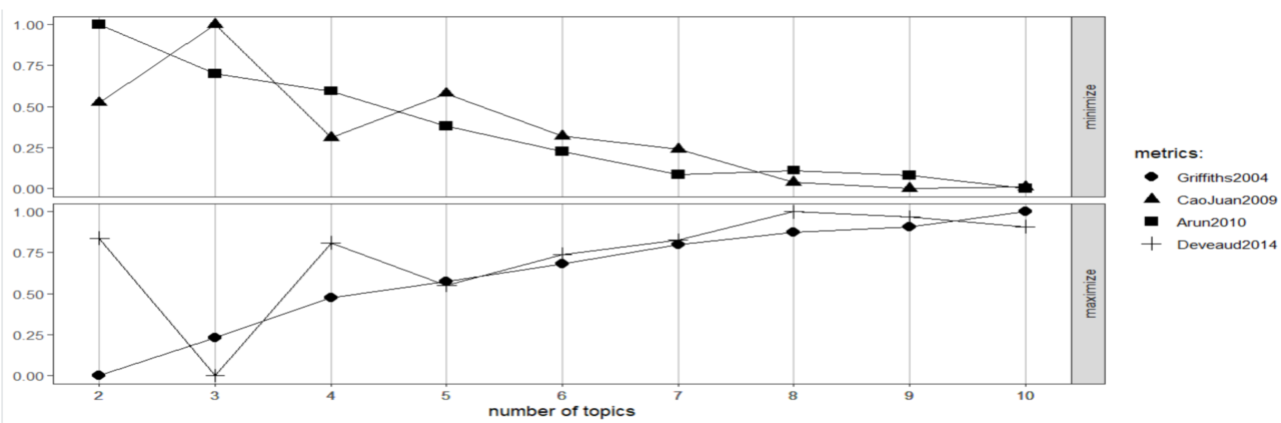
- seed=10 - נקבע את הסיד כדי לקבל חלוקה עקבית שווה
- alpha=0.1 - נקבע את האלפא כמספר קטן בשביל במסמך מסווים ישתייך לנושא מסוים ולא להרבה נושאים. כלומר בחירה באלפא 0.1 על מנת שבסבירות גבוהה האלגוריתם ישייך לטופיק מסויים.
- "method = "Gibbs - נבחר את שיטת החלוקה ל GIBBS

מספר קבוצות :

ננסה לבצע חלוקה של 4, 8 , 9

K=8 - נראה כי חלוקה ל-8 אשכולות מייצרת נקודות קיצון (מינ/מקס) במספר אלגוריתמים ולכן זה מחזק את הבחירה במספר זה.

גרף תומך החלטה:



ביקורות שליליות:

פרמטרים :

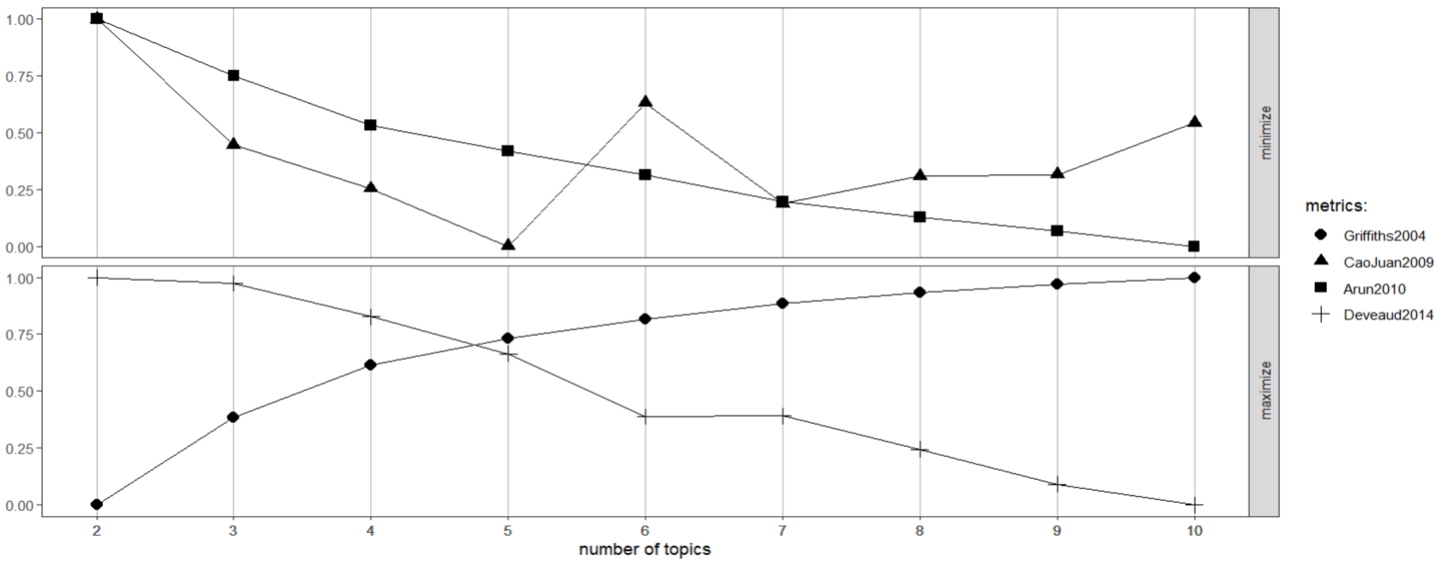
- seed=10 - נקבע את הסיד כדי לקבל חלוקה עקבית שווה
- alpha=0.1 - נקבע את האלפא כמספר קטן בשביל במסמך מסווים ישתייך לנושא מסוים ולא להרבה נושאים. כלומר בחירה באלפא 0.1 על מנת שבסבירות גבוהה האלגוריתם ישייך לטופיק מסויים.
- "method = "Gibbs - נבחר את שיטת החלוקה ל GIBBS

מספר קבוצות :

ננסה לבצע מספר חלוקות של 3, 5 , 7

K=5 - נראה כי חלוקה ל-5 אשכולות מייצרת נקודות קיצון (מינ/מקס) במספר אלגוריתמים ולכן זה מחזק את הבחירה במספר זה

גרף תומך החלטה:



שאלה 4

ביקורות חיוביות:

עשרת המילים המרכזיות בכל קבוצה:

```
> termsByTopic_positive
      Topic 1  Topic 2 Topic 3    Topic 4  Topic 5     Topic 6  Topic 7 Topic 8
 [1,] "home"   "arriv" "room"     "stay"   "great"     "walk"   "apart" "apt"
 [2,] "area"   "leav"  "good"     "boston" "locat"     "boston" "get"   "jamaica"
 [3,] "much"   "get"   "bed"      "hous"   "stay"      "stay"   "good"  "sara"
 [4,] "trip"   "room"  "nice"     "make"   "place"     "place"  "stay"  "plain"
 [5,] "space"  "late"  "clean"    "host"   "apart"     "good"   "night" "part"
 [6,] "feel"   "time"  "bathroom" "great"  "recommend" "great"  "peopl" "ray"
 [7,] "full"   "give"  "stay"     "home"   "boston"    "apart"  "door"  "suppli"
 [8,] "long"   "find"  "realli"   "good"   "host"      "locat"  "great" "clean"
 [9,] "access" "day"   "comfort"  "feel"   "clean"     "nice"   "air"   "jim"
[10,] "like"   "meet"  "need"     "place"  "comfort"   "also"   "one"   "paul"
>
```

```
1. Living Experience and Accessibility :
 home area much trip space feel full long access like

2. Availability and Timing :
 arriv leav get room late time give find day meet

3. Comfort and Quality of Stay :
 room good bed nice clean bathroom stay realli comfort need

4. Experiences and Hosting in Boston :
 stay boston hous make host great home good feel place

5. Location and Recommended Hosting :
 great locat stay place apart recommend boston host clean comfort

6. Positive Atmosphere Stay :
 walk boston stay place good great apart locat nice also

7. Service and Social Interactions :
 apart get good stay night peopl door great air one

8. Apartments and Clean Environment :
 apt jamaica sara plain part ray suppli clean jim paul
```

<u>ביקורות שליליות:</u>

עשרת המילים המרכזיות בכל קבוצה:

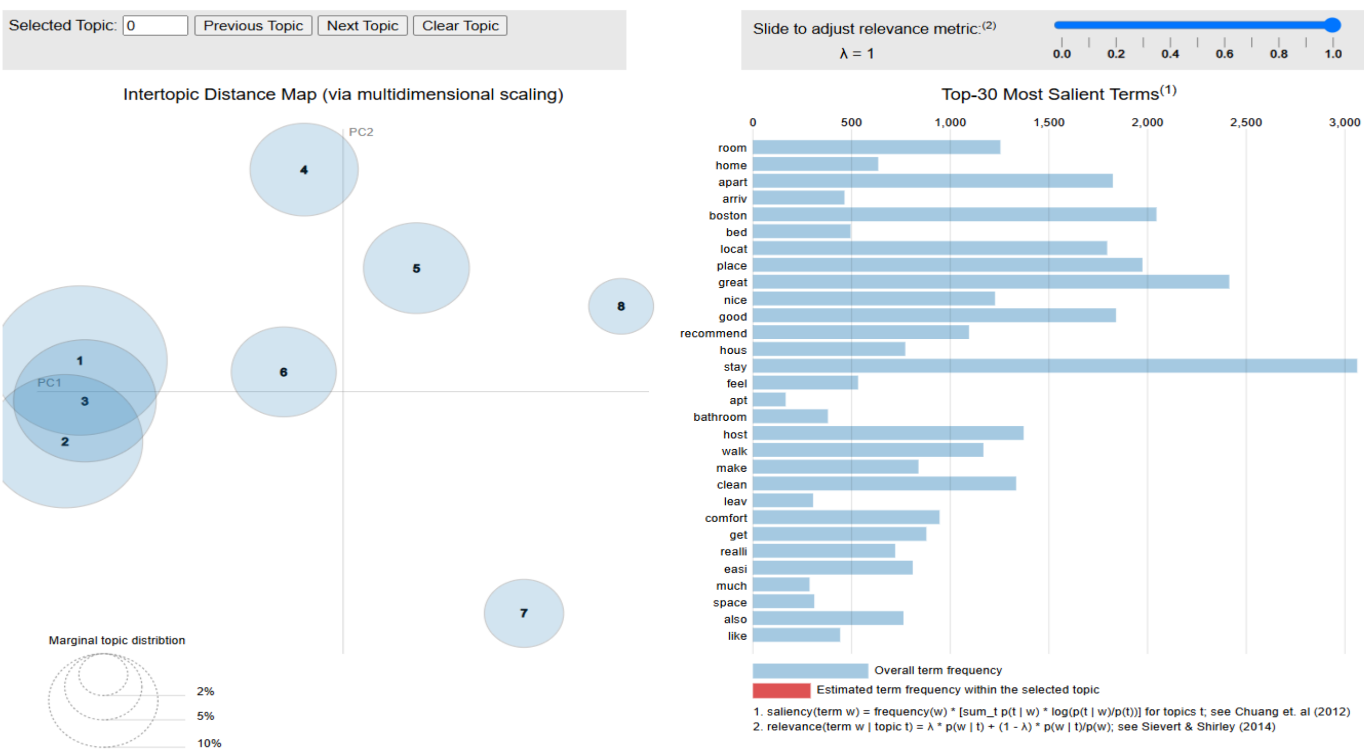
```
> termsByTopic_negative
      Topic 1    Topic 2   Topic 3   Topic 4  Topic 5
 [1,] "room"     "locat"   "door"    "place"  "arriv"
 [2,] "apart"    "boston"  "terribl" "leav"   "find"
 [3,] "clean"    "take"    "email"   "good"   "dirti"
 [4,] "nice"     "great"   "recent"  "like"   "stay"
 [5,] "quit"     "park"    "robert"  "night"  "can"
 [6,] "bathroom" "close"   "also"    "stay"   "extrem"
 [7,] "back"     "just"    "casa"    "bed"    "also"
 [8,] "friend"   "never"   "con"     "host"   "feel"
 [9,] "just"     "receiv"  "entir"   "pay"    "idea"
[10,] "much"     "avail"   "home"    "time"   "inform"
>
```

```
1. Room and Accommodation Quality :
 room apart clean nice quit bathroom back friend just much

2. Location and Accessibility :
 locat boston take great park close just never receiv avail

3. Negative Experiences and Complaints :
 door terribl email recent robert also casa con entir home

4. General Stay and Host Feedback :
 place leav good like night stay bed host pay time

5. Arrival and Communication Issues :
 arriv find dirti stay can extrem also feel idea inform
```

שאלה 5

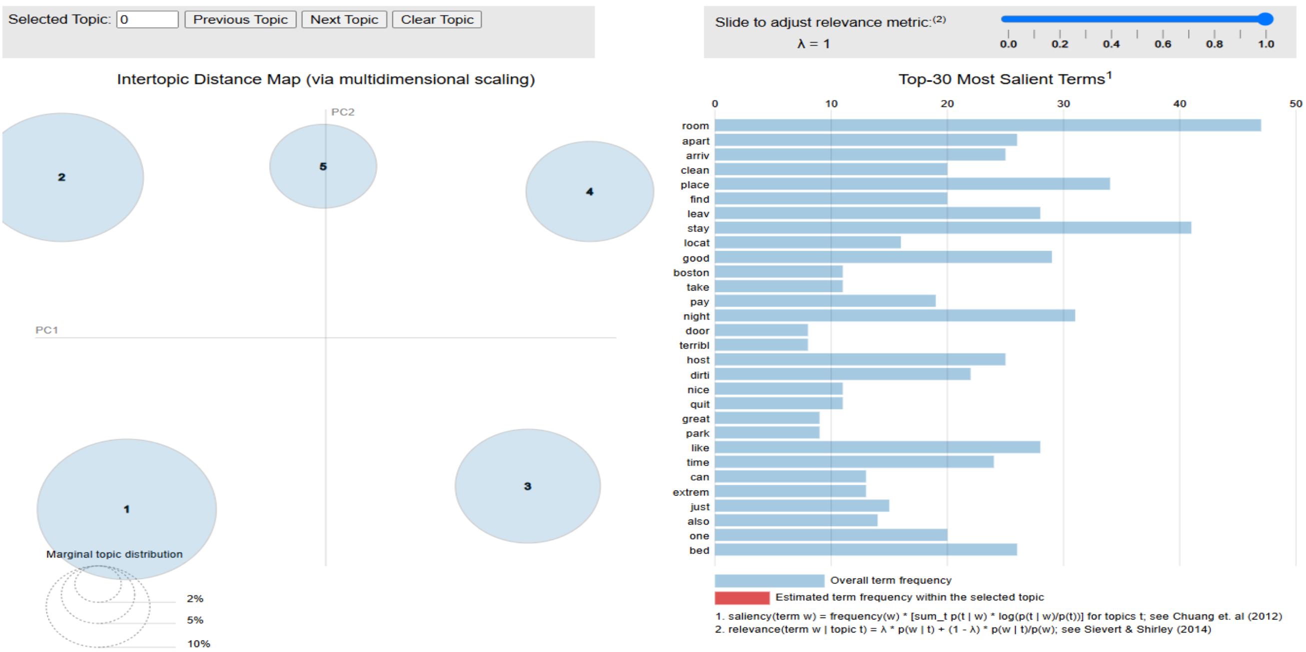
צילומי מסך של שתי הויזואליזציות:

ביקורות חיובית:



ביניהם, אך קיימים שלושה טופיקים שבהם יש חפיפה מסוימת. למרות זאת, החלוקה של שאר הטופיקים מצוינת ומעידה על חלוקה מרשימה

ביקורות שליליות:



האם החלוקות טובות לדעתכם – ביקורת שלילית:

החלוקה של הטופיקים מראה תוצאות מרשימות, עם הפרדה ברורה. המרחקים המשמעותיים בין הטופיקים והעובדה שאף אחד אינו חופף או עולה על נושא אחר מעידים על חלוקה מעולה

```r
# Using libraries
library(stringi)
library(textstem)
library(tm)
library(qdap)
library(sentimentr)
library(wordcloud)
library(ggthemes)
library(lexicon)
library(topicmodels)
# set working directory
setwd("C:/Users/niv/Desktop/Text mining/EX3")

#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.1~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
# Set Options
options(stringsAsFactors=FALSE) #don't treat strings as factors (categories)
Sys.setlocale('LC_ALL','C')

# read csv
airbnb.df<-read.csv('bos_airbnb_1k.csv')
head(airbnb.df)

#text coloum into new data frame
comments.df<-airbnb.df[,'comments']
head(comments.df)

# Sentiment analysis
airbnb.df$sentiment <- sentiment_by(comments.df,question.weight=0)$ave_sentiment
head(airbnb.df$sentiment)
```

```r
table(sign(airbnb.df$sentiment))

#Dividing positive and negative comments
negative.df<-airbnb.df[airbnb.df$sentiment<0,'comments']
positive.df<-airbnb.df[airbnb.df$sentiment>0,'comments']

#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.2~~~~~~~~~~~~~~~~~~~~~~~~~~~

#clean.data function

negative.df<- gsub("\n"," ",negative.df)
positive.df<-gsub("\n"," ",positive.df)

negative.df<- textclean::replace_non_ascii(negative.df,replacement = "")
positive.df<- textclean::replace_non_ascii(positive.df,replacement = "")

clean.data <- function(text){
  myCorpus <- VCorpus(VectorSource(text))
  myCorpus <- tm_map(myCorpus, content_transformer(tolower))# convert to lower case
  myCorpus <- tm_map(myCorpus, removeWords, stopwords("english")) #remove stopwords
  myCorpus <- tm_map(myCorpus, removeNumbers)# remove numbers
  myCorpus <- tm_map(myCorpus, removePunctuation)# remove punctuation
  myCorpus <- tm_map(myCorpus, stripWhitespace)# remove extra whitespace
  myCorpus <- tm_map(myCorpus, content_transformer(lemmatize_strings))
  myCorpus <- tm_map(myCorpus, stemDocument)
  return(myCorpus)
}

positive.Corpus<-clean.data(positive.df)
```

```
negative.Corpus<-clean.data(negative.df)

#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.3~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

dtm_positive <- DocumentTermMatrix(positive.Corpus)
dtm_negative <- DocumentTermMatrix(negative.Corpus)

# Choose number of topics
library(ldatuning)

# positive
result_positive <- FindTopicsNumber(
  dtm_positive,
  topics = seq(from = 1, to = 10, by = 1),
  metrics = c("Griffiths2004", "CaoJuan2009", "Arun2010", "Deveaud2014"),
  method = "Gibbs", #VEM/Gibbs
  control = list(seed = 10,alpha=0.1),
  mc.cores = 2L,
  verbose = TRUE
)
FindTopicsNumber_plot(result_positive)

# negative
result_negative <- FindTopicsNumber(
  dtm_negative,
  topics = seq(from = 1, to = 10, by = 1),
  metrics = c("Griffiths2004", "CaoJuan2009", "Arun2010", "Deveaud2014"),
  method = "Gibbs", #VEM/Gibbs
  control = list(seed = 10,alpha=0.1),
```

```
  mc.cores = 2L,
  verbose = TRUE
)
FindTopicsNumber_plot(result_negative)


##run LDA with 8 topics (positive)
lda_positive <- LDA(dtm_positive, k = 8, control=list(seed=10,alpha=0.1),method = "Gibbs") #lower alpha, we assume documents contain fewer topics
termsByTopic_positive <- terms(lda_positive, 10)
termsByTopic_positive


##run LDA with 5 topics (negative)
lda_negative <- LDA(dtm_negative, k = 5, control=list(seed=10,alpha=0.1),method = "Gibbs") #lower alpha, we assume documents contain fewer topics
termsByTopic_negative <- terms(lda_negative, 10)
termsByTopic_negative


topics_positive <- topics(lda_positive)
topics_positive  #assignment of term to a topic
table(topics_positive) #number of terms in each topic


topics_negative <- topics(lda_negative)
topics_negative  #assignment of term to a topic
table(topics_negative) #number of terms in each topic



#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.4~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
# The most common words in each group


# positive
```

```r
# Define custom titles for each topic
topic_titles <- c(
  "1. Living Experience and Accessibility",
  "2. Availability and Timing",
  "3. Comfort and Quality of Stay",
  "4. Experiences and Hosting in Boston",
  "5. Location and Recommended Hosting",
  "6. Positive Atmosphere Stay",
  "7. Service and Social Interactions",
  "8. Apartments and Clean Environment"
)

# Print topic titles with terms
for (i in 1:ncol(termsByTopic_positive)) {
  cat(topic_titles[i], ":\n", termsByTopic_positive[, i], "\n\n")
}

# negative
# Define custom titles for each topic
topic_titles <- c(
  "1. Room and Accommodation Quality",
  "2. Location and Accessibility",
  "3. Negative Experiences and Complaints",
  "4. General Stay and Host Feedback",
  "5. Arrival and Communication Issues"
)

# Print topic titles with terms
```

```
for (i in 1:ncol(termsByTopic_negative)) {
  cat(topic_titles[i], ":\n", termsByTopic_negative[, i], "\n\n")
}

#~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~Q.5~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
# interactive visualization
library(LDAvis)
library(servr)
# Convert the output of a topicmodels Latent Dirichlet Allocation to JSON
# for use with LDAvis
topicmodels2LDAvis <- function(x, ...){
  post <- topicmodels::posterior(x)
  if (ncol(post[["topics"]]) < 3) stop("The model must contain > 2 topics")
  mat <- x@wordassignments
  LDAvis::createJSON(
    phi = post[["terms"]],
    theta = post[["topics"]],
    vocab = colnames(post[["terms"]]),
    doc.length = slam::row_sums(mat, na.rm = TRUE),
    term.frequency = slam::col_sums(mat, na.rm = TRUE)
  )
}
# Apply the function
#positive
Jason_positive <- topicmodels2LDAvis(lda_positive)
serVis(Jason_positive)
#negative
Jason_negative <- topicmodels2LDAvis(lda_negative)
serVis(Jason_negative)
```